

# SkillForge AI — Product Requirements Document

**Problem Statement:** SIH26101 | **Organization:** MoSPI — Data Informatics & Innovation Division (DIID) **Version:** 4.0 | **Date:** August 2026 | **Prepared for:** Smart India Hackathon 2026

**Revision note:** v4.0 finalizes the core feature set (competency assessment, skill-gap analysis, recommendation/learning-path engine, MCQ generator, adaptive quiz engine, and a RAG-grounded AI tutor), adopts a deterministic-scoring design principle for explainability, and updates the architecture and data model to support retrieval-augmented generation.

## 1. Document Basics

**Purpose:** Define the finalized, build-ready scope of SkillForge AI — what it does, for whom, and exactly how each AI component works, precisely enough that “how does that actually compute” always has a real answer.

**Owner:** Abdul Rehman, Team Lead **Status:** Final pre-build draft **Source of truth:** Official Problem Statement 26101 (MoSPI/DIID); mentor review notes; team feature specification

## 2. Problems and Goals

### 2.1 The Problem

Officials need targeted upskilling in a fast-evolving technical domain layered onto deep statistical domain work. Existing platforms (iGOT Karmayogi, generic enterprise LMS) tell an officer what courses exist; none measure, with evidence, what an officer specifically doesn’t yet know before a course is chosen.

### 2.2 What Existing Platforms Do (researched)

| Platform               | Competency source                                                                                               | Recommendation logic                                        |
|------------------------|-----------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------|
| iGOT Karmayogi (FRAC)  | Broad role-to-framework mapping; 2026 “Trust-Based Assessment” validates mastery <i>after</i> course completion | Tag-matching; completion-driven                             |
| Generic enterprise LMS | Self-reported profile fields or completion history                                                              | Tag-matching between profile and course metadata            |
| SkillForge AI          | Diagnostic assessment producing a deterministic, evidence-based score                                           | Hybrid semantic + rule-based ranking, with stated reasoning |

### 2.3 Goals

- Measure competency with evidence, computed deterministically — not asserted by an LLM
- Make the framework role-specific, not one checklist for everyone
- Classify and prioritize gaps (Critical/High/Medium/Low), each with a stated reason
- Recommend an ordered learning pathway, not a flat course list
- Ground all learner-facing AI assistance in the actual uploaded material, not open-ended generation
- Close the loop: reassess after learning, and visibly update the score

### 2.4 Novelty and Differentiation

Three structural differences from every platform reviewed:

1. **Diagnostic-before-training, not validation-after-training** — competency is measured before a course is assigned, not confirmed after one is completed.
2. **Role-specific target vectors** — each role has its own weighted required-proficiency profile, not a shared generic checklist.
3. **A living, recalibrating profile** — every assessment updates the score; the system’s picture of an officer never goes stale.

### 2.5 Core Design Principle (adopted this revision)

**The LLM is responsible for language and content generation, never for deciding a number.** Competency scores, gap sizes, and recommendation rankings are computed by deterministic rules and formulas over structured evidence (assessment

answers, difficulty weights, role targets). The LLM’s job is limited to: generating assessment questions, writing plain-language reasoning for a gap or recommendation, and powering the grounded AI tutor. This is what makes every score in the product explainable and defensible under questioning, and it directly avoids the failure mode of “the AI decided” being the only answer available.

2.6 Non-Goals

- Replacing iGOT as a course host
- Full HRIS/SSO integration, multilingual UI, video learning content, real-time adaptive branching quizzes
- Custom-trained ML models — all intelligence is prompting + retrieval + deterministic scoring over LLM APIs
- Predictive workforce forecasting (acknowledged in the wider spec as P2 — architecture-only, not built)

3. Users and Scope

3.1 Personas

**Learner/Officer** — builds a profile, takes diagnostic and post-learning assessments, sees competency scores and prioritized gaps, follows a generated learning path, uses the AI tutor while learning.

**Trainer** — uploads learning material, generates and reviews MCQs, configures assessments, reviews learner performance and topic-level weak spots.

**Administrator** — monitors organization-wide competency distribution, department/role-level gaps, training effectiveness, and critical skill shortages.

3.2 Finalized Feature Scope

| Feature                                                  | Priority                 | Notes                                                                                                            |
|----------------------------------------------------------|--------------------------|------------------------------------------------------------------------------------------------------------------|
| Intelligent Officer Profile                              | P0                       | Structured fields; NLP-assisted extraction where useful, scoring stays deterministic                             |
| Competency Framework & Skill Knowledge Base              | P0                       | Four-domain taxonomy; role → required competency mapping; skill prerequisites (e.g. Python → Data Analysis → ML) |
| AI-Based Competency Assessment (D)                       | P0                       | Diagnostic generation via LLM; scoring via deterministic formula                                                 |
| Automated Skill-Gap Analysis (E)                         | P0                       | Deterministic comparison + severity classification + LLM-written reasoning                                       |
| Recommendation & Learning Path Engine (F)                | P0                       | Hybrid semantic + rule-based ranking; ordered pathway, not a flat list                                           |
| AI-Powered MCQ/Quiz Generator (J)                        | P0                       | RAG-grounded generation from uploaded material; trainer review before publish                                    |
| Assessment & Adaptive Quiz Engine (K)                    | P0 core / P1 stretch     | Difficulty-weighted scoring is P0; live branching adaptivity is stretch                                          |
| AI Tutor / RAG Assistant (L)                             | P0 (confirmed in scope)  | Grounded in approved learning material; retrieval before generation                                              |
| iGOT/NSSTA adapter (seeded dataset)                      | P0                       | Clearly labelled mock data; live API is future work                                                              |
| Admin Workforce Intelligence                             | P0 (org/department view) | Predictive analytics explicitly P2/out of scope                                                                  |
| Video transcription, SSO, background-worker infra polish | P2                       | Architecture-ready, not required for demo                                                                        |

4. Features and Requirements

#### 4.1 Intelligent Officer Profile

- Captures designation, department, current assignment, education, experience, previous training, current/target role
- Maintains an auditable update history
- **Acceptance criteria:** a profile update never silently overwrites prior evidence — history is retained

#### 4.2 Competency Framework & Skill Knowledge Base

- Four-domain taxonomy (Statistical, Technical, Digital Governance, Behavioural/Managerial), seeded from the official problem statement
- Role → required competency + required proficiency level mapping (the role-specific target vector)
- Skill prerequisite relationships stored explicitly (e.g., Python → Data Analysis → Machine Learning) — used by the Learning Path Engine to sequence recommendations
- **Acceptance criteria:** two roles produce visibly different required profiles; prerequisite chains are queryable, not implicit

#### 4.3 AI-Based Competency Assessment (D)

- Diagnostic assessment generated from the officer's role and required competencies
- Score computed by a deterministic formula combining: assessment correctness (weighted by question difficulty), prior training evidence, and previous assessment history — never an LLM-assigned number
- Every score displays its evidence (which answers, which prior training contributed)
- Recalculates after each new assessment or completed learning item
- **Acceptance criteria:** the same evidence always produces the same score (deterministic, testable); the score's evidence is visible on demand, not hidden

#### 4.4 Automated Skill-Gap Analysis (E)

- Compares current competency score against role-required level (plus department priority and optional target-role level)
- Computes gap size and classifies severity: **Critical / High / Medium / Low**
- LLM generates a one-sentence, specific reason per gap — never a generic statement
- **Acceptance criteria:** severity classification follows a stated, consistent threshold rule; every gap has a reason string

#### 4.5 Recommendation & Learning Path Engine (F)

- Matches gaps to iGOT and NSSTA TPAC entries using semantic similarity (embeddings) plus rule-based factors: role relevance, gap severity, prerequisites, difficulty, prior learning, department priority
- Produces a transparent ranking score, not a black-box order
- Generates an **ordered learning pathway** (e.g., a 6-8 week sequence), not a flat course list
- Every recommendation shows a “Why this was recommended?” explanation
- **Acceptance criteria:** no course appears without a visible reason and rank; the pathway respects stored prerequisite order

#### 4.6 Document & Learning Content Processing

- Accepts PDF, DOCX, PPTX (video/audio transcription is P2, not built for the demo)
- Extracts and chunks text, generates embeddings, stores vectors in PostgreSQL via pgvector
- Original files go to object storage (Supabase Storage), never inside the relational database
- Tracks processing status and failure states visibly

#### 4.7 AI-Powered MCQ & Quiz Generator (J)

- Trainer selects question count, topic, difficulty; generation is RAG-grounded (retrieval from the document's own chunks/embeddings), not free-form
- Each question includes answer choices, correct answer, explanation, and source context (which chunk it came from)
- Trainer reviews/edits before publishing — nothing goes live unreviewed
- **Acceptance criteria:** every generated question can be traced to a specific source chunk

#### 4.8 Assessment & Adaptive Quiz Engine (K)

- Diagnostic, standard, and post-learning assessment types
- Automatic evaluation with instant feedback

- Difficulty-weighted scoring (P0); live adaptive difficulty branching is a stretch goal only if time permits
- Results map to specific competencies and feed back into Section 4.3’s recalculation
- **Acceptance criteria:** an assessment result always updates the linked competency score, visibly, without manual intervention

4.9 AI Tutor / RAG Assistant (L) — confirmed in scope

- Course-aware assistant for learner questions, grounded in approved learning material only
- Retrieves relevant chunks (via the same pgvector store as Section 4.6) before generating a response
- Explains concepts calibrated to the learner’s current competency level
- Supports follow-up actions: “give an example,” “summarize this,” “quiz me” (the last one hands off to Section 4.8)
- **Acceptance criteria:** every substantive answer is grounded in retrieved content, not open-ended generation; the tutor states when a question falls outside the uploaded material rather than guessing

4.10 Dashboards

**Learner:** overall readiness score, competency-by-domain visualization (Caliper gauges), prioritized gaps, learning path, learning hours/completions/scores, target-role readiness, next recommended action, AI tutor access.

**Trainer:** material management, question bank review, assessment publishing, learner performance view, topics with poor performance flagged.

**Administrator:** organization-wide competency distribution, department/role-level gaps, training effectiveness, critical shortages, drill-down from organization → department → role → skill.

4.11 Non-Functional Requirements

| Category    | Requirement                                                                                                                   |
|-------------|-------------------------------------------------------------------------------------------------------------------------------|
| Security    | RBAC across Learner/Trainer/Admin; protected API routes with authorization checks on every sensitive operation; audit logging |
| Performance | Diagnostic generation + scoring ≤ 30s; MCQ generation ≤ 30s; dashboard load ≤ 2s                                              |
| Reliability | Explicit error/fallback states on LLM or embedding failure                                                                    |
| Compliance  | Aligned with the Digital Personal Data Protection Act, 2023                                                                   |

5. Design and Technical Details

5.1 Architecture Summary

Next.js 14 (App Router), API routes as backend, PostgreSQL + Prisma, NextAuth.js for auth, Gemini via a server-side AI adapter, hosted on Vercel + managed PostgreSQL/storage. Named AI/intelligence components:

- **Competency Engine** — role mapping, deterministic scoring from assessment evidence
- **Skill Gap Engine** — deterministic comparison, severity classification, prioritization
- **Recommendation Engine** — hybrid semantic (pgvector) + rule-based ranking
- **Learning Path Engine** — orders recommendations using the prerequisite graph + gap priority
- **Assessment Engine** — diagnostic/adaptive question delivery, difficulty-weighted scoring
- **MCQ Generator** — RAG-grounded generation (chunking + embeddings + LLM), trainer-reviewed
- **AI Tutor** — embeddings + retrieval + grounded LLM response

**New infrastructure this revision:** pgvector extension on PostgreSQL for embeddings; object storage (Supabase Storage) for original documents; a background worker/queue for embedding generation and heavy document processing, so uploads don’t block the request thread.

5.2 Core Data Model (entities)

- **User** — profile, role, department, education, experience, training history
- **Role** — name, description, required competencies (target vector)

- **Competency** — skill, domain, level, description, prerequisites
- **RoleCompetency** — role → competency → required level
- **DepartmentPriority** — department → skill → priority/future demand
- **UserCompetency** — user → competency → current score → evidence reference
- **Course** — source (iGOT/NSSTA), title, skills, level, prerequisites, duration
- **Recommendation** — user, course, score, reasons, status
- **Assessment** — type, competency mapping, difficulty, questions
- **Question** — stem, options, answer, explanation, source chunk reference
- **QuizAttempt** — user, assessment, answers, score, timestamp
- **LearningProgress** — course, enrolment, completion, time spent
- **Document** — owner, storage location, type, processing status
- **DocumentChunk** — document reference, text, embedding vector
- **AuditLog** — actor, action, resource, timestamp, metadata

5.3 Golden Demo Flow (the actual live walkthrough)

1. Officer logs in, profile is complete
2. System maps role → required competencies
3. Diagnostic assessment measures current competency
4. Skill Gap Engine identifies and prioritizes gaps (Critical/High/Medium/Low)
5. Recommendation Engine surfaces relevant iGOT + NSSTA entries
6. Learning Path Engine produces an ordered pathway
7. Officer opens the AI Tutor, asks a grounded question about course material
8. Trainer uploads a document, generates and reviews an MCQ set
9. Officer completes the quiz, gets instant feedback
10. Competency score updates from the new evidence, visibly
11. Platform recommends the next step
12. Administrator opens Workforce Intelligence, sees department-level gaps

5.4 Onboarding Flow

Unchanged in principle from v3: minimum profile (Designation, Department, Job Role) → first diagnostic instead of a self-rating form → preliminary gap view → progressive diagnostics across sessions → routed to the single highest-priority next action, never a full dashboard on first use.

5.5 Visual Design Direction

“The Caliper” — retained, unchanged: measurement-instrument aesthetic, grotesk sans + mono numerals, cool paper tone, light/dark toggle, no gamification. The gap-severity classification (Critical/High/Medium/Low) maps onto the Caliper’s existing red/teal tinting logic without needing a new visual system.

6. Planning and Risks

6.1 Build Phases

| Phase | Focus                                                                        |
|-------|------------------------------------------------------------------------------|
| 1     | Auth/RBAC, data model, competency taxonomy + prerequisite graph seed data    |
| 2     | Competency Engine (deterministic scoring) + Diagnostic Assessment generation |
| 3     | Skill Gap Engine (severity classification + reasoning)                       |
| 4     | Document pipeline: upload → chunk → embed (pgvector) → object storage        |
| 5     | MCQ Generator (RAG-grounded) + Assessment Engine                             |

|   |                                                                   |
|---|-------------------------------------------------------------------|
|   | scoring/feedback loop                                             |
| 6 | Recommendation + Learning Path Engine (seeded iGOT/NSSTA dataset) |
| 7 | AI Tutor (retrieval + grounded response + follow-up actions)      |
| 8 | Dashboards (Learner/Trainer/Admin), visual pass, demo rehearsal   |

6.2 Risks and Mitigations

| Risk                                                                        | Mitigation                                                                                                                                                                |
|-----------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| RAG infra (embeddings, pgvector, chunking) eats disproportionate build time | Build document pipeline early (Phase 4) as a dependency for both J and L; reuse the same pipeline for both features rather than building twice                            |
| Judges question “how is the score computed”                                 | Section 2.5’s deterministic-scoring principle is the standing answer — rehearse it explicitly                                                                             |
| AI Tutor gives ungrounded/hallucinated answers live                         | Retrieval-then-generate only; tutor explicitly states when a question is outside the uploaded material                                                                    |
| LLM API failure during live demo                                            | Pre-computed diagnostic + pre-generated tutor exchange cached as fallback                                                                                                 |
| Scope still too large for remaining time                                    | Build phases are ordered so Phases 1-3 + 6 alone already demonstrate the full closed loop (Section 5.3, steps 1-6, 10-12) even if J/L slip — protect that core path first |

6.3 Assumptions and Dependencies

- Gemini API access remains available through build and demo
- pgvector extension is available on the chosen managed PostgreSQL provider
- Target vectors and course dataset cover 3-5 representative roles, illustrative not exhaustive
- Evaluation centers on the closed-loop journey in Section 5.3 working live, end-to-end, for one officer